

Assignment by Vinoth kumar C M

1. Student Registration: Java CallableStatement program to call registerStudent() and insert student name, age, and department.

Student Registration System — Procedure in **SQL**

```
CREATE procedure registerStudent(  
    IN p_name VARCHAR(100),  
    IN p_age INT,  
    IN p_department VARCHAR(100)  
)  
BEGIN  
    Insert into student(name, age, department)  
    VALUES(p_name, p_age, p_department);  
END;
```

Java

```
import java.sql.*;  
  
public class StudentRegistration {  
    public static void main(String[] args) {  
        String url = "jdbc:mysql://localhost:3306/college";  
        String user = "root";  
        String password = "*****";  
  
        try {  
            Connection con = DriverManager.getConnection(url, user, password);  
            CallableStatement cs =  
                con.prepareCall("{call registerStudent(?, ?, ?)}");  
            cs.setString(1, "qqqq");  
            cs.setInt(2, 25);  
            cs.setString(3, "IT");  
            cs.execute();  
            System.out.println("Student registered successfully");  
        }  
    }  
}
```

```

        cs.close();

        con.close();
    } catch (Exception e) {
        System.out.println(e);
    }
}
}
}

```

2. Employee Salary: Java CallableStatement program to call getEmployeeSalary() with employee ID and retrieve salary using ResultSet.

Employee Salary Search — Procedure Created in **Java**

```

import java.sql.*;

public class EmployeeSalarySearch {
    public static void main(String[] args) {
        String url = "jdbc:mysql://localhost:3306/company";
        String user = "root";
        String password = "*****";
        try {
            Connection con = DriverManager.getConnection(url, user, password);
            Statement st = con.createStatement();
            st.executeUpdate(
                "CREATE PROCEDURE getEmployeeSalary(IN p_id INT) " +
                "BEGIN " +
                "SELECT salary FROM employee WHERE id = p_id; " +
                "END"
            );
            CallableStatement cs =
                con.prepareCall("{call getEmployeeSalary(?)}");
            cs.setInt(1, 103);

```

```

        ResultSet rs = cs.executeQuery();

        if (rs.next()) {

            System.out.println("Salary: " + rs.getDouble("salary"));

        } else {

            System.out.println("Employee not found");

        }

        rs.close();

        cs.close();

        st.close();

        con.close();

    } catch (Exception e) {

        System.out.println(e);

    }

}
}

```

3.Bank Transfer: Java CallableStatement program to call transferAmount() and transfer ₹5,000 between two accounts.

Bank Account Transfer — Procedure in **SQL**

```

CREATE PROCEDURE transferAmount(

    IN p_sender INT,

    IN p_receiver INT,

    IN p_amount DECIMAL(10,2)

)

BEGIN

    UPDATE account

    SET balance = balance - p_amount

    WHERE account_no = p_sender;

    UPDATE account

```

```
SET balance = balance + p_amount  
WHERE account_no = p_receiver;  
END;
```

Java:

```
import java.sql.*;  
  
public class BankTransfer {  
    public static void main(String[] args) {  
        String url = "jdbc:mysql://localhost:3306/bank";  
        String user = "root";  
        String password = "*****";  
  
        try {  
            Connection con = DriverManager.getConnection(url, user, password);  
            CallableStatement cs =  
                con.prepareCall("{call transferAmount(?, ?, ?)}");  
            cs.setInt(1, 7007);  
            cs.setInt(2, 7777);  
            cs.setDouble(3, 7700);  
            cs.execute();  
            System.out.println("Transfer successful");  
            cs.close();  
            con.close();  
        } catch (Exception e) {  
            System.out.println(e);  
        }  
    }  
}
```